

Find and Fix: Does the Data Really Show a Cause?

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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| 1. 3 | 6. the mean score of the few-books group = 72, — |
| 2. the greatest number of rescues = 7, — | 7. the difference in damage = 170, — |
| 3. the difference in rescues = 5, — | 8. the mean of the sleep-schedule group = 85, the mean of the comparison group = 77, the difference between the two means = 8, — |
| 4. the corrected percentage = 75, — | |
| 5. the mean score of the many-books group = 84, — | |
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What is verified

10 of 17 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.SP.A.1 – problems 1, 2, 3, 5, 6, 7, 8

8.SP.A.4 – problem 4

Difficulty 1–5 of 5 across 17 tagged checks.

Common wrong answers

2. *If they got 48: read across to the ice-cream column and reported its largest figure instead*
4. *If they got 22.5: divided by the whole sample of 200 instead of the 60 instrument players*
7. *If they got 24: subtracted the numbers of firefighters instead of the damage figures*
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1. Six summer weeks: find the mean number of beach rescues.

Solution: Add the six rescue counts and divide by six:

$$\frac{1 + 2 + 4 + 5 + 3 + 3}{6} = \frac{18}{6}$$

2. Five towns: give the greatest number of rescues, and diagnose Ola's conclusion.

(a) **Solution:** Read along the rescues row — not the ice-cream row above it. The largest entry is Cardel's. 7

(b) **Diagnosis (instructor-judged).** A correct response says Ola has found an association and called it a cause. The two rows do rise together, but a table of totals cannot show that one produced the other, and there is no plausible mechanism by which eating ice cream would put a swimmer in danger. Accept any wording that separates a pattern from a cause. Correcting the number without addressing the claim is not a diagnosis.

3. Five towns: the Cardel-Ashby difference, and a third factor.

(a) **Solution:** Cardel had 7 rescues and Ashby had 2, so the difference is $7 - 2$. 5

(b) **Third factor (instructor-judged).** The natural answer is hot weather. A hot spell sends more people out to buy ice cream *and* more people into the sea, so both rows climb together without either causing the other. Town size or a busy tourist season are acceptable alternatives, provided the response explains the effect on *both* rows. Naming a factor that would move only one row does not earn full credit.

4. Jack wrote $\frac{45}{200} \times 100 = 22.5\%$. Correct it and say what went wrong.

(a) **Solution:** The question asks about the instrument players, so the denominator is the size of that group, 60, not the whole sample: $\frac{45}{60} \times 100 = 0.75 \times 100$. 75

(b) **Diagnosis (instructor-judged).** A correct response says Jack used the wrong denominator: he divided by all 200 students when only the 60 instrument players were being asked about. Accept any wording that names the right denominator. Note the size check — the answer must be a percentage of a group in which 45 out of 60 succeeded, so it cannot possibly be under a quarter.

5. Book survey: mean of the more-than-100-books group, and the headline.

(a) **Solution:** $\frac{78 + 85 + 82 + 91 + 84}{5} = \frac{420}{5}$. 84

(b) **Judgement (instructor-judged).** A correct response says the headline is not justified. The survey observed students as they already were; nobody changed how many books anyone owned, so all it shows is that the two groups differ. Homes with many books usually differ in other ways too — income, reading habits, time spent with an adult reading — and any of those could produce the gap. Accept any wording that rejects the causal claim and points at the observational nature of the survey.

6. Book survey: mean of the fewer-than-100-books group, and what evidence would be needed.

(a) **Solution:** $\frac{70 + 75 + 68 + 77 + 70}{5} = \frac{360}{5}$. 72

(b) **Proposal (instructor-judged).** The strongest answer describes an experiment: take a group of similar students, decide *at random* which of them receive extra books, keep everything else the same, and compare their later scores. A response that instead names a

confounding factor — family income, reading at home — and says it would have to be held equal also earns full credit. Repeating that the survey is not enough, with no proposal at all, earns partial credit.

7. Fire services: Ely's damage above Alton's, and Rita's reasoning.

(a) Solution: Work in the damage row: $210 - 40$. Subtracting the firefighter counts instead would give 24, which is a count of people rather than an amount of money. 170

(b) Diagnosis (instructor-judged). A correct response either turns the direction around — towns with more fires need more firefighters, so the damage comes first — or names town size as the factor behind both: bigger towns have more buildings, so more and larger fires, so both more firefighters and more damage. Sending more firefighters to a fire reduces damage; it cannot create it. Accept either argument as long as it explains the pattern in the table.

8. Two studies: find both means and their difference, then say which can support a cause.

Solution:

$$\frac{82 + 88 + 79 + 91 + 85}{5} = \frac{425}{5}, \quad \frac{74 + 80 + 71 + 83 + 77}{5} = \frac{385}{5}$$

The two means are 85 and 77, and the gap between them is the difference of those two numbers.

85	77	difference = 8
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Judgement (instructor-judged). A correct response chooses Study 1 and explains the reason: the students were put into the two groups *at random*, so the groups started out alike and the sleep schedule was the only planned difference between them. That is what lets the eight-mark gap be attributed to the schedule. Study 2 only recorded what students had already chosen for themselves, so the more motivated students could easily account for both the app and the higher scores. Accept any wording that contrasts random assignment with students choosing for themselves.